DESIGN AND ANALYSIS OF ALGORITHM

LAB EXPERIMENT -7

NAME: INBARASU

ROLL NO: CH.SC.U4CSE24242

1. JOB SCHEDULLING CODE:

```
  GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
struct Job {
    char id;
    int deadline;
    int profit;
};
int compare(const void *a, const void *b) {
    struct Job *j1 = (struct Job *)a;
    struct Job *j2 = (struct Job *)b;
    return j2->profit - j1->profit;
}
void jobSequencing(struct Job jobs[], int n) {

    qsort(jobs, n, sizeof(struct Job), compare);
    int result[n];
    int slot[n];
    for(int i = 0; i < n; i++)
        slot[i] = 0;
    for(int i = 0; i < n; i++) {

        for(int j = jobs[i].deadline - 1; j >= 0; j--) {

            if(slot[j] == 0) {
                result[j] = i;
                slot[j] = 1;
                break;
            }
        }
    }

    printf("Selected jobs: ");
    for(int i = 0; i < n; i++) {
        if(slot[i])
            printf("%c ", jobs[result[i]].id);
    }
}
int main() {
    printf("\nroll no : ch.sc.u4cse24215");
    printf("\nname : hemal");
    struct Job jobs[] = {
        {'A', 2, 100},
        {'B', 1, 19},
        {'C', 2, 27},
        {'D', 1, 25},
        {'E', 3, 15}
    };
    int n = sizeof(jobs) / sizeof(jobs[0]);
    jobSequencing(jobs, n);
    return 0;
}
```

OUTPUT:

```
hemal@Admin:~$ nano job.c
hemal@Admin:~$ gcc job.c
hemal@Admin:~$ gcc job.c -o job
hemal@Admin:~$ ./job

roll no : ch.sc.u4cse24215
name : hemalSelected jobs: C A E hemal@Admin:~$
```

INTERPRETATION:

Job Scheduling with deadlines is a greedy algorithm used to schedule jobs in a way that maximizes total profit while completing each job before its deadline.
It works by sorting jobs in decreasing order of profit and assigning them to the latest available time slot before their deadline.